1. Write a shell script to generate mark-sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

Windows PowerShell ISE (x86)

File Edit View Tools Debug Add-ons Help

Untitled1.ps1*

```
$m1 = Read-Host "Enter marks of Subject 1"
$m2 = Read-Host "Enter marks of Subject 2"
$m3 = Read-Host "Enter marks of Subject 3"

$total = [int]$m1 + [int]$m2 + [int]$m3
$percentage = $total / 3

Write-Host "Total Marks = $total"
Write-Host "Percentage = $percentage %"

if ($percentage -ge 60) {
    Write-Host "Class: First Class"
}
elseif ($percentage -ge 50) {
    Write-Host "Class: Second Class"
}
elseif ($percentage -ge 40) {
    Write-Host "Class: Pass"
}
else {
    Write-Host "Class: Fail"
}
```

```
if ($percentage -ge 60) {
    Write-Host "Class: First Class"
}
elseif ($percentage -ge 50) {
    Write-Host "Class: Second Class"
}
elseif ($percentage -ge 40) {
    Write-Host "Class: Pass"
}
else {
    Write-Host "Class: Fail"
}
Enter marks of Subject 1: 88
Enter marks of Subject 2: 76
Enter marks of Subject 3: 71
Total Marks = 235
Percentage = 78.3333333333333 %
Class: First Class

PS C:\Windows\system32>
```

Commands

Modules: All Refresh

Name:

A:
Add-AppProvisionedSharedPackageContainer
Add-AppSharedPackageContainer
Add-AppvClientConnectionGroup
Add-AppvClientPackage
Add-AppvPublishingServer
Add-AppxPackage
Add-AppxProvisionedPackage
Add-AppxVolume
Add-BCDataCacheExtension
Add-BitLockerKeyProtector
Add-BitsFile
Add-CertificateEnrollmentPolicyServer
Add-Computer
Add-Content
Add-DnsClientDohServerAddress
Add-DnsClientNrptRule
Add-DtcClusterTMMapping
Add-EtwTraceProvider
Add-History
Add-InitiatorIdToMaskingSet
Add-JobTrigger
Add-KdsRootKey
Add-Member
Add-MpPreference
Add-NetEventNetworkAdapter
Add-NetEventPacketCaptureProvider
Add-NetEventProvider
Add-NetEventVFPProvider

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2.Write a menu driven shell script which will print the following menu and execute the given task.

Windows PowerShell ISE (x86)

File Edit View Tools Debug Add-ons Help

Untitled1.ps1* Untitled2.ps1*

```
Write-Host "1. Display calendar of current month"
Write-Host "2. Display today's date and time"
Write-Host "3. Display users currently logged in"
Write-Host "4. Display terminal number"

$ch = Read-Host "Enter your choice"

switch ($ch) {
    1 { Get-Date -Format "MMMM yyyy" }
    2 { Get-Date }
    3 { query user }
    4 { $Host.Name }
    default { Write-Host "Invalid choice" }
}
```

```
PS C:\Windows\system32> Write-Host "1. Display calendar of current month"
Write-Host "2. Display today's date and time"
Write-Host "3. Display users currently logged in"
Write-Host "4. Display terminal number"

$ch = Read-Host "Enter your choice"

switch ($ch) {
    1 { Get-Date -Format "MMMM yyyy" }
    2 { Get-Date }
    3 { query user }
    4 { $Host.Name }
    default { Write-Host "Invalid choice" }
}
1. Display calendar of current month
2. Display today's date and time
3. Display users currently logged in
4. Display terminal number
Enter your choice: 1
January 2026

PS C:\Windows\system32> |
```

Commands

Modules: All Refresh

Name:

A:
Add-AppProvisionedSharedPackageContainer
Add-AppSharedPackageContainer
Add-AppvClientConnectionGroup
Add-AppvClientPackage
Add-AppvPublishingServer
Add-AppxPackage
Add-AppxProvisionedPackage
Add-AppxVolume
Add-BCDataCacheExtension
Add-BitLockerKeyProtector
Add-BitsFile
Add-CertificateEnrollmentPolicyServer
Add-Computer
Add-Content
Add-DnsClientDohServerAddress
Add-DnsClientNrptRule
Add-DtcClusterTMMapping
Add-EtwTraceProvider
Add-History
Add-InitiatorIdToMaskingSet
Add-JobTrigger
Add-KdsRootKey
Add-Member
Add-MpPreference
Add-NetEventNetworkAdapter

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3. Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

Windows PowerShell ISE (x86)

File Edit View Tools Debug Add-ons Help

Untitled1.ps1* Untitled2.ps1* Untitled3.ps1*

```
$n = Read-Host "Enter number of terms"

$a = 0
$b = 1

Write-Host "Fibonacci series:"
for ($i=1; $i -le $n; $i++) {
    Write-Host -NoNewline "$a "
    $c = $a + $b
    $a = $b
    $b = $c
}
Write-Host ""
```

```
4. Display terminal number
Enter your choice: 1
January 2026

PS C:\Windows\system32> $n = Read-Host "Enter number of terms"

$a = 0
$b = 1

Write-Host "Fibonacci series:"
for ($i=1; $i -le $n; $i++) {
    Write-Host -NoNewline "$a "
    $c = $a + $b
    $a = $b
    $b = $c
}
Write-Host ""
Enter number of terms: 20
Fibonacci series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

PS C:\Windows\system32> |
```

Commands

Modules: All Refresh

Name:

A:
Add-AppProvisionedSharedPackageContainer
Add-AppSharedPackageContainer
Add-AppvClientConnectionGroup
Add-AppvClientPackage
Add-AppvPublishingServer
Add-AppxPackage
Add-AppxProvisionedPackage
Add-AppxVolume
Add-BCDataCacheExtension
Add-BitLockerKeyProtector
Add-BitsFile
Add-CertificateEnrollmentPolicyServer
Add-Computer
Add-Content
Add-DnsClientDohServerAddress
Add-DnsClientNrptRule
Add-DtcClusterTMMapping
Add-EtwTraceProvider
Add-History
Add-InitiatorIdToMaskingSet
Add-JobTrigger
Add-KdsRootKey
Add-Member
Add-MpPreference
Add-NetEventNetworkAdapter

Run Insert Copy

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4. Write a shell script which will accept a number b and display first n prime numbers as output

Windows PowerShell ISE (x86)

File Edit View Tools Debug Add-ons Help

Untitled1.ps1* Untitled2.ps1* Untitled3.ps1* Untitled4.ps1*

```
$n = Read-Host "Enter value of n"

$count = 0
$num = 2

while ($count -lt $n) {
    $isPrime = $true

    for ($i=2; $i -le [math]::Sqrt($num); $i++) {
        if ($num % $i -eq 0) {
            $isPrime = $false
            break
        }
    }

    if ($isPrime) {
        Write-Host -NoNewline "$num "
        $count++
    }
    $num++
}
```

```
while ($count -lt $n) {
    $isPrime = $true

    for ($i=2; $i -le [math]::Sqrt($num); $i++) {
        if ($num % $i -eq 0) {
            $isPrime = $false
            break
        }
    }

    if ($isPrime) {
        Write-Host -NoNewline "$num "
        $count++
    }
    $num++
}
Write-Host ""
Enter value of n: 20
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71

PS C:\Windows\system32>
```

Commands

Modules: All Refresh

Name:

A:
Add-AppProvisionedSharedPackageContainer
Add-AppSharedPackageContainer
Add-AppvClientConnectionGroup
Add-AppvClientPackage
Add-AppvPublishingServer
Add-AppxPackage
Add-AppxProvisionedPackage
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Add-DnsClientDohServerAddress
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Add-DtcClusterTMMapping
Add-EtwTraceProvider
Add-History
Add-InitiatorIdToMaskingSet
Add-JobTrigger
Add-KdsRootKey
Add-Member
Add-MpPreference
Add-NetEventNetworkAdapter

Run Insert Copy

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5. Write menu driven program for file handling activity
Creation of file
Write content in the file
Upend file content

Windows PowerShell ISE (x86)

File Edit View Tools Debug Add-ons Help

Untitled1.ps1* Untitled2.ps1* Untitled3.ps1* Untitled4.ps1* Untitled5.ps1*

```
Write-Host "1. Create file"
Write-Host "2. Write content in file"
Write-Host "3. Append file content"
Write-Host "4. Delete file content"

$ch = Read-Host "Enter choice"
$fname = Read-Host "Enter filename"

switch ($ch) {
    1 { New-Item $fname -ItemType File }
    2 {
        $content = Read-Host "Enter content"
        Set-Content $fname $content
    }
    3 {
        $content = Read-Host "Enter content to append"
        Add-Content $fname $content
    }
    4 { Clear-Content $fname }
    default { Write-Host "Invalid choice" }
}
```

```
$ch = Read-Host "Enter choice"
$fname = Read-Host "Enter filename"

switch ($ch) {
    1 { New-Item $fname -ItemType File }
    2 {
        $content = Read-Host "Enter content"
        Set-Content $fname $content
    }
    3 {
        $content = Read-Host "Enter content to append"
        Add-Content $fname $content
    }
    4 { Clear-Content $fname }
    default { Write-Host "Invalid choice" }
}
1. Create file
2. Write content in file
3. Append file content
4. Delete file content
Enter choice: 1
```

Commands

Modules: All Refresh

Name:

A:
Add-AppProvisionedSharedPackageContainer
Add-AppSharedPackageContainer
Add-AppvClientConnectionGroup
Add-AppvClientPackage
Add-AppvPublishingServer
Add-AppxPackage
Add-AppxProvisionedPackage
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Add-BCDataCacheExtension
Add-BitLockerKeyProtector
Add-BitsFile
Add-CertificateEnrollmentPolicyServer
Add-Computer
Add-Content
Add-DnsClientDohServerAddress
Add-DnsClientNrptRule
Add-DtcClusterTMMapping
Add-EtwTraceProvider
Add-History
Add-InitiatorIdToMaskingSet
Add-JobTrigger
Add-KdsRootKey
Add-Member
Add-MpPreference
Add-NetEventNetworkAdapter

Run Insert Copy

Running script / selection. Press Ctrl+Break to stop. Press Ctrl+B to break into debugger.

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